4.1 Three Dimensional Coordinate Systems

In a 3D coordinate system, a point is represented by 3 coordinates using three coordinate axes perpendicular to each other at origin namely X, Y, Z axis

2D transformations can be represented by 3 x 3 matrices using homogenous coordinates, so 3D transformations can be represented by 4 x 4 matrices using homogeneous coordinate representations of points.

Thus instead of representing a point as (x , y ,z) , we represent if as (x , y , z, H), where two these quadruples represent the same point it one is a non zero multiple of the other the quadruple (0,0,0,0) is not allowed.

A standard representation of a point (x , y ,z ,H) with H not zero is given by (x/H , y/H, z/H, 1).

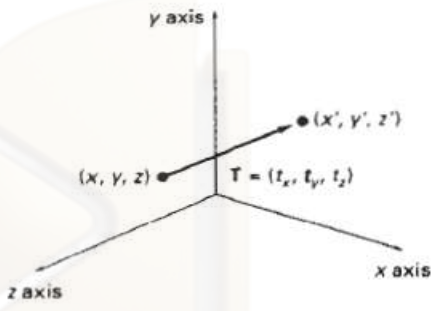
Transforming the point to this form is called homogenizing.

4.2 Three Dimensional Transformations

Translation:
A point is translated from position P=(x,y,z) to position P' = (x',y',z') with the matrix operation

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

or P' = T.P



Parameters t_x , t_y ,t_z specify translation distances for the coordinate directions x ,y and z.

This matrix representation is equivalent to three equations:

$x' = x + t_x$ $\quad$ $y' = y + t_y$ $\quad$ $z' = z + t_z$

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Scaling:
Scaling changes size of an object and repositions the object relative to the coordinate origin.

If transformation parameters are not all equal then figure gets distorted

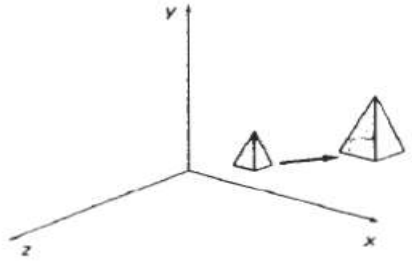
So we can preserve the original shape of an object with uniform scaling($s_x = s_y = s_z$)

Matrix expression for scaling transformation of a position P = (x,y,z) relative to the coordinate origin can be written as :

or P' = S . P

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

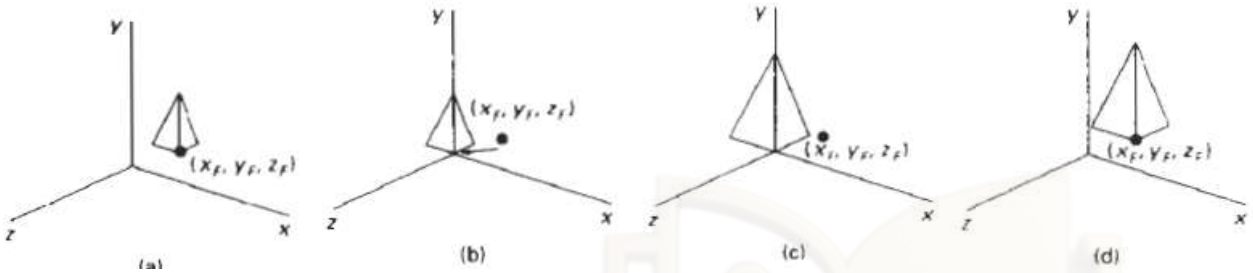
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Scaling with respect to a selected fixed point (x_f, y_f, z_f) can be represented with:

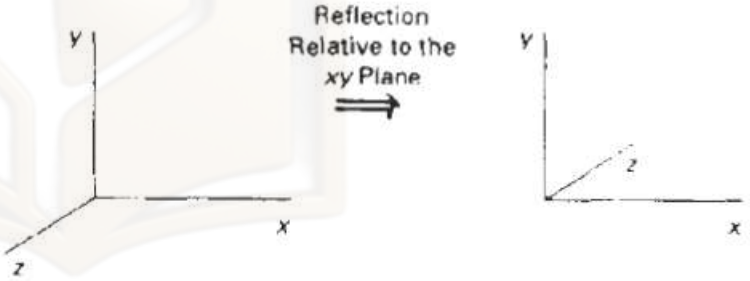
i. Translate fixed point to the origin
ii. Scale object relative to the coordinate origin
iii. Translate fixed point back to its original position

or $T(x_f, y_f, z_f).\ S(s_x, s_y, s_z).T(-x_f, -y_f, -z_f)$



Reflection:
Reflections with respect to a plane are equivalent to 180* rotations in four dimensional space.

$$RFz = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



This transformation changes the sign of the z coordinates, leaving the x and y coordinate values unchanged.

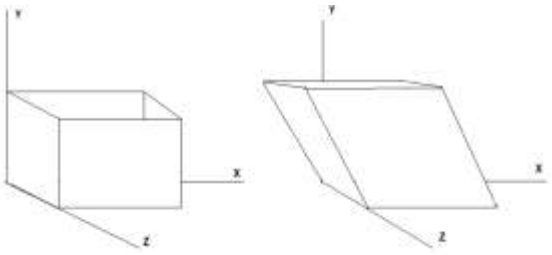
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Transformation matrices for inverting x and y values are defined similarly, as reflections relative to the yz plane and xz plane.

Shearing:
Shearing transformations are used to modify object shapes.
e.g. shears relative to the z axis:

$$SHz = \begin{pmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

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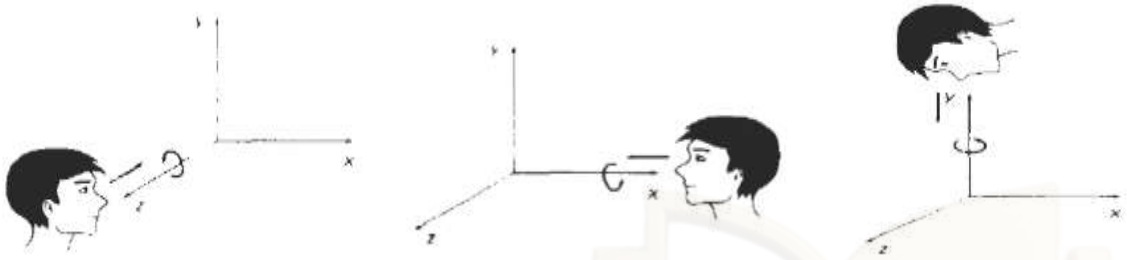
where parameters a and b assume any real values.

It alters the x and y coordinate values by an amount that is proportional to the z value while leaving the z coordinate unchanged.

Shearing matrices for x and y axis can be obtained similarly.

Rotation:

Designate the axis of rotation about which the object is to be rotated and the amount of angular rotation .



Axes that are parallel to the coordinate axes are easy to handle.

Coordinate axes Rotations:

2D z-axis rotation equations are easily extended to 3D:

x' = xcosθ - ysinθ y' = xsinθ + ycosθ z' = z ……(i)

3D z-axis rotation equations are expressed in homogenous coordinate form as

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$



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or,

P' = Rz (θ) . P

Cyclic permutation of the coordinate parameters x , y and z are used to get transformation equations for rotations about the other two coordinates

x → y → z → so,

substituting permutations in (i) for an x axis rotation we get,

y' = ycosθ - zsinθ z' = ysinθ + zcosθ x' = x

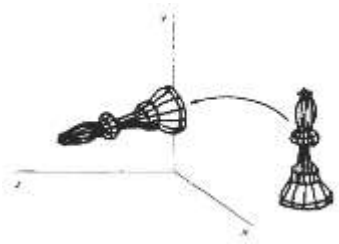
3D x-axis rotation equations in homogenous coordinate form as

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

or,

P' = Rx (θ) . P

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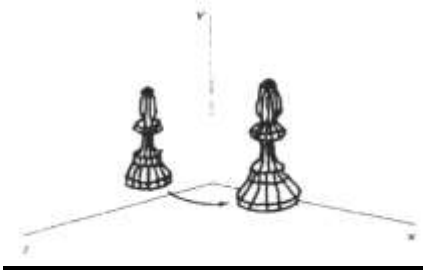
substituting permutations in (i) for a y axis rotation we get,

$z' = z\cos\theta - x\sin\theta$ $\quad x' = z\sin\theta + x\cos\theta$ $\quad y' = y$

3D y-axis rotation equations are expressed in homogenous coordinate form as

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

or, $P' = Ry(\theta) \cdot P$

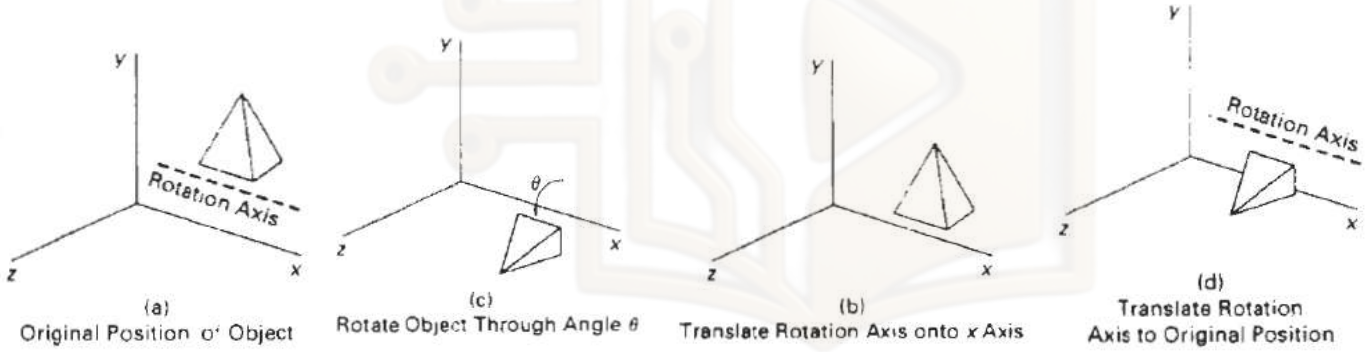


Rotation about an axis parallel to one of the coordinate axes :

Steps:

i. Translate object so that rotation axis coincides with the parallel coordinate axis.
ii. Perform specified rotation about that axis
iii. Translate object back to it's original position.

ie. $P' = T^{-1} \cdot R_x(\theta) \cdot T \cdot P$



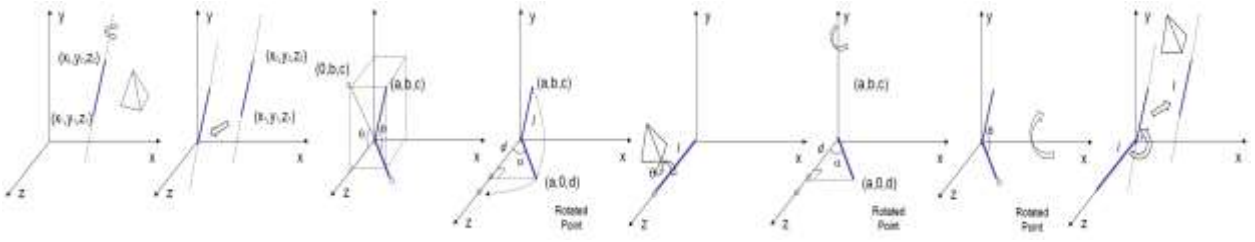
Rotation about an arbitrary axis :

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An arbitrary axis in space passing thru point (x_0, y_0, z_0) with direction cosines (c_x, c_y, c_z)

Steps required for rotation about this axis by some angle 0 are:

i. Translate so that point(x_0, y_0, z_0) is at the origin of the coordinate system
ii. Perform rotations to make axis of rotation coincident with the z axis
iii. Rotate about the z axis by the angle θ.
iv. Perform the inverse of the combined rotation transformation
v. Perform the inverse of the translation.



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$$CM = \begin{bmatrix} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & y_1 \\ 0 & 0 & 1 & z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & 0 & \sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi & 0 & 0 \\ \sin\phi & \cos\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ \sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Making an arbitrary axis passing thru origin coincident with one of the coordinate axes requires two successive rotations about the other two coordinate axes.

NOTE:

- **See class notes for rotation about arbitrary axis and rotation about arbitrary plane**

Reflection thru an arbitrary plane :

General reflection matrices cause reflection thru x = 0 y = 0 z = 0 coordinate planes respectively.

Often it is necessary to reflect an object thru a plane other than one of these, which is obtained with the help of a series of transformations (composition).

i. translate a known point P that lies in the reflection plane to the origin of the coordinate system
ii. rotate the normal vector to the reflection plane at the origin until it is coincident with the z axis
this makes the reflection plane the z = 0 coordinate plane

iii. after applying the above transformation to the object reflect the object thru the z = 0 coordinate plane.
i.e.

$$RFz = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

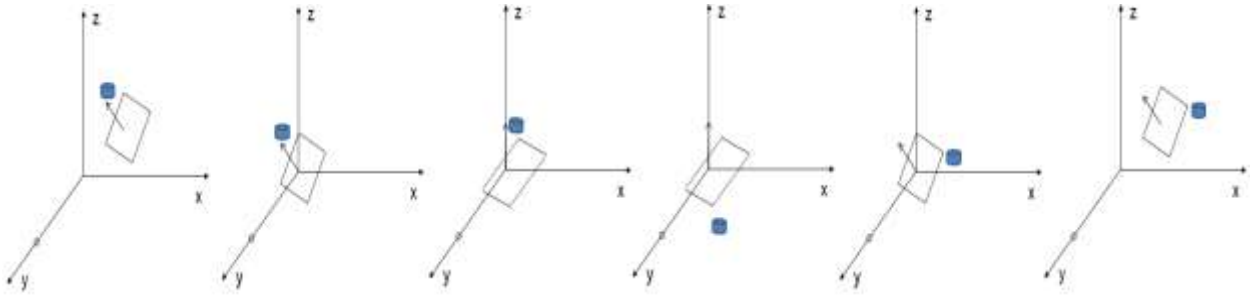
iv. Perform the inverse transformation to those given above to achieve the desired result.

So the general transformation matrix is:

$$M(\theta) = T^{-1} \cdot R_x^{-1}(\theta) \cdot R_y^{-1}(\theta) \cdot Rflct_z(\theta) \cdot R_y(\theta) \cdot R_x(\theta) \cdot T$$

$$CM = \begin{bmatrix} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & y_1 \\ 0 & 0 & 1 & z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & 0 & \sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha & 0 \\ 0 & 1 & 0 & 0 \\ \sin\alpha & 0 & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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NOTE:

- **See class notes for rotation about arbitrary axis and rotation about arbitrary plane**

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